

PIRABU KANNAN. U

AI/ML Engineer | LLM & Agentic AI | MLOps | Digital Twins | Aerospace & Autonomous Systems

pirabukannanum19@gmail.com | +91-90800 22667 | Chennai, India

SUMMARY

AI/ML and Digital Twin Engineer with 5+ years of experience developing intelligent software and engineering systems across aerospace, advanced manufacturing, embedded systems, and automation. Strong hands-on experience with Python, AI/ML, FastAPI, REST APIs, LLM API integration, Agentic AI, Docker, Linux, cloud platforms, real-time data integration, and Digital Twin development.

Experienced in integrating simulation, engineering data, AI/ML workflows, and real-time industrial systems to support condition monitoring, prediction, diagnosis, optimization, and intelligent decision-making. Developed an aero-engine Digital Twin for a GTRE-sponsored project and currently developing Python-based AI/ML Digital Twin software for advanced manufacturing process monitoring and automation.

Strong systems background spanning AI software, backend services, embedded control, sensors, industrial communication, robotics, and automation, with a demonstrated ability to work across multidisciplinary engineering environments and rapidly adopt new technologies.

CORE TECHNICAL SKILLS

AI / Machine Learning

- Python | Machine Learning | AI/ML Model Development | Predictive Analytics | Condition Monitoring | Fault Diagnosis | Prognostics | Intelligent Optimization | Decision Support

LLM & Intelligent Applications

- LLM API Integration | Agentic AI | AI Application Development | AI-Assisted Workflows | Prompt/Model Integration

Software & Backend

- Python | FastAPI | REST APIs | C | Embedded C | Git | Linux/SSH | Docker

Cloud & Infrastructure

- AWS | Microsoft Azure | Google Cloud Platform | Containerized Applications | Development & Deployment Environments

Data & Industrial Integration

- MQTT | OPC UA | PLC Integration | Industrial IoT | Real-Time Data Processing | Sensor Data Integration | LabVIEW

Digital Twin & Simulation

- Digital Twin Development | Engineering-System Modeling | Simulation Integration | Real-Time Monitoring | Predictive Maintenance | Anomaly/Fault Analysis | Unity3D | MATLAB

Embedded & Autonomous Systems

- Microcontrollers | Embedded Software | PCB Design & Debugging | Robotics | Automation | Sensor Integration | CAN | SPI | I2C | USART | RS232 | RS485 | USB

Engineering Tools

- ANSYS | COMSOL (Basic) | SolidWorks | AutoCAD | MATLAB | LabVIEW | Unity3D

PROFESSIONAL EXPERIENCE

Senior Project Officer, Indian Institute of Technology Madras

Jan 2026 – Present

- Developing Python-based Digital Twin and AI/ML software for advanced manufacturing process monitoring, prediction, and automation.
- Developing intelligent workflows for integrating engineering/process data with AI/ML models to support real-time monitoring and decision-making.
- Working on software architectures combining AI/ML, process modeling, data processing, and automation for complex engineering systems.

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PROFESSIONAL EXPERIENCE

- Developing capabilities for condition assessment, intelligent optimization, and automated process support.

Project Officer, Indian Institute of Technology Madras

Jan 2025 – Dec 2025

- Developed Python-based Digital Twin and AI/ML workflows for process-data analysis, condition monitoring, prediction, and intelligent optimization.
- Integrated engineering-system data processing with AI/ML workflows to support monitoring, analysis, and decision support.
- Developed software-oriented workflows for transforming engineering data into actionable insights for intelligent machine-system operation.
- Worked across Python, AI/ML, Digital Twin architecture, data processing, and engineering automation.

Project Associate, Indian Institute of Technology Madras

Apr 2022 – Dec 2024

Aero-Engine Digital Twin — GTRE Sponsored Project

- Developed a Digital Twin for an aero-engine system in a project sponsored by the Gas Turbine Research Establishment (GTRE), Bangalore.
- Integrated simulation, data-processing, and AI/ML workflows for engineering-system analysis and decision support.
- Worked on software workflows connecting engineering-system behavior with data-driven analysis.
- Applied multidisciplinary engineering knowledge spanning aerospace propulsion, AI/ML, simulation, data processing, and Digital Twin development.

Associate Engineer – Mechatronics,

Jul 2021 – Mar 2022

System Control Technology Solutions Pvt. Ltd.

- Worked as part of an embedded software team, developing and debugging microcontroller-based control applications.
- Programmed microcontrollers and supported embedded control-system development and integration.
- Designed and analyzed PCBs and performed hardware/software troubleshooting.
- Integrated multiple subsystems and conducted system-level functional testing.
- Developed practical experience working at the interface of software, electronics, sensors, control, and physical systems.

RESEARCH EXPERIENCE

AI/ML based Digital Twin & Intelligent Systems Development

- Developed frameworks for online condition monitoring, fault diagnosis, prognostics, and intelligent optimization of machine systems.
- Backend & AI: Python, FastAPI, REST APIs, AI/ML, LLM API Integration, Agentic AI, C, Embedded C.
- Development & Infrastructure: Git, Linux/SSH, Docker, AWS, Azure, GCP.
- Industrial & Automation Integration: MQTT, OPC UA, PLC, LabVIEW, MATLAB, Unity3D.
- Integrated engineering-system data, simulation, AI/ML workflows, and automation interfaces for intelligent system development.

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RESEARCH PROJECTS

Aero-Engine Digital Twin

- GTRE-Sponsored Research Project
- Developed a Digital Twin for an aero-engine system.
- Integrated simulation, data processing and AI/ML workflows for engineering analysis and decision support.

Digital Twin for CNC Milling Machine

- Developed Digital Twin concepts for machine-system monitoring and intelligent analysis.

Digital Twin for CNC Drilling Machine

- Developed Digital Twin-based approaches for engineering-system monitoring and analysis.

Three-Axis Surveillance Pedestal

- Worked on a three-axis surveillance pedestal system involving mechanical/electronic system integration.

Pan-Tilt Pedestal with Thermal Monitoring

- Developed a pan-tilt pedestal system incorporating thermal monitoring capabilities.

Autonomous Oil Spill remover Systems

- Developed an Autonomous Oil Spill Removal Robot, demonstrating experience in robotics, automation and autonomous-system concepts.

AI/ML for Manufacturing Surfaces

- Developed AI/ML models for micro-machined patterned surfaces.
- Developed AI/ML models for machining surfaces.

Sensor & Instrumentation Projects

- Worked on sensor development for heavy-metal detection.
- Developed a portable potentiostat.

Embedded Control & Automation

- Worked with microcontrollers including Arduino, Raspberry Pi, ATmega2560, ATmega128 and AT90CAN128.
- Worked with communication protocols including MQTT, USART, SPI, SSI, I2C, CAN, USB, RS232 and RS485.
- Experience in robotics, automation, PCB troubleshooting, relay logic and 3D printing.

EDUCATION

Master of Science (Mechanical Engineering)

Indian Institute of Technology Madras, Chennai

CGPA: 9.00/10 | 2026

Relevant Coursework:

- Aerospace Propulsion
- Artificial Intelligence in Manufacturing
- Industrial Internet of Things
- Sensors for Intelligent Manufacturing and Condition Monitoring
- Prognostics and Health Management for Machine Tools

Bachelor of Engineering – Mechatronics

Kongu Engineering College, Erode

CGPA: 9.33/10 | 2021

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ACADEMIC ACHIEVEMENT

- Top 5% in Academic Excellence Award during undergraduate program.
- Second Place — National Innovation Challenge, Tishk International University, Erbil, Iraq.

SELECTED ACHIEVEMENTS

- Second Place — National Innovation Challenge, Tishk International University, Iraq.
- Second Place — Sanitation Design Challenge, IEEE Kochi.
- Second Place — National Institute of Technology, Trichy project/presentation competition.
- Top 5% academic performance during undergraduate engineering.

WORKSHOPS & ADDITIONAL EXPERIENCE

Drone Workshop

IEEE Kongu Engineering College Student Branch

- Design, analysis and control of basic drone systems.

Tracker Workshop

Robotics Club, Kongu Engineering College

- Electronics and microcontroller fundamentals.